

ORIGINAL ARTICLE

Erika Schagatay · Boris Holm

Effects of water and ambient air temperatures on human diving bradycardia

Accepted: 1 August 1995

Abstract Upon apnoeic face immersion, humans develop a diving response resembling that found in diving mammals. There have been contradictory reports regarding the influence of water temperature on the magnitude of the resulting bradycardia. This study examined the influence of both water and ambient air temperatures on human diving bradycardia. A group of 23 volunteers performed three series of apnoeic episodes after 60-min exposure to air at temperatures of 10, 20 or 30°C. Oral and skin temperatures were measured during this exposure and during the subsequent test on 5 subjects. At 20°C air temperature oral and skin temperatures were measured on 10 subjects. Heart rate (HR) was recorded for the 23 subjects during apnoea in air and apnoea with the face immersed in water of 10, 20 or 30°C, at each air temperature. We found that both air and water temperatures had significant effects on immersion bradycardia, but in opposite directions. Face immersion in cold water after exposure to a high ambient air temperature induced the most pronounced bradycardia. We further observed that exposure to different ambient air temperatures resulted in different patterns of HR response to water temperature. The range in which the response was positively correlated to water temperature differed at 30°C ambient air from that at 10 and 20°C ambient air. We concluded from these studies that human bradycardia resulting from apnoeic face immersion is inversely proportional to water temperature within a range which is determined by the ambient air temperature. Thus, the interval in which the response to cold stimulation varies with temperature, would appear to be determined by the ambient temperature before stimulation.

Key words Diving response · Apnoea · Face immersion · Bradycardia · Temperature

Introduction

The diving response was first described in diving mammals (Scholander 1940; Andersen 1966), and has been shown to occur in forced, trained and free dives in a variety of species (for review see Kooyman 1989). The response permits extensive diving by limiting the use of oxygen. Oxygen conservation is achieved mainly by selective vasoconstriction and heart rate (HR) reduction (bradycardia). The blood flow has been shown to be diverted away from organs that can function anaerobically, but is maintained to organs sensitive to asphyxia, such as the heart and the brain (Andersen 1966), and to any hard working muscles (Butler and Woakes 1987). The diving response is initiated when the head of a diving mammal is submerged. The two principal sensory inputs have been described as facial neural activity and cessation of respiratory movements (Dykes 1974).

It has been shown that a response similar to that of habitually diving mammals has been obtained in humans during breath-hold (apnoeic) diving, or when the face is immersed in cold water and the breath is held (Furedy and Morrison 1983; review Elsner and Gooden 1983). It has been reported that apnoea alone reduces HR by about half as much as apnoea plus facial cooling (Hurwitz and Furedy 1986) and the response appears to be induced by the same mechanisms as those in diving mammals. Since it has been found that dry cold stimulation of the face results in a bradycardia of the same magnitude as facial immersion, the involvement of facial cold-receptors in initiating the response is clear in humans (Kawakami et al. 1967). It has been shown that chilling of the forehead results in a more pronounced bradycardia than that which is obtained

E. Schagatay (✉) · B. Holm
Department of Animal Physiology, University of Lund,
S-223 62 Lund, Sweden

¹Formerly K.E. Schuitema

when other facial areas of similar size are chilled (Schuitema and Holm 1988). This has indicated that the receptors involved in triggering the response are located mainly in the forehead, or are more densely distributed in the forehead than in other parts of the face (Schuitema and Holm 1988). It has been shown that after 25 to 30 s of apnoea and cold stimulation, the bradycardia is fully established (Kawakami et al. 1967; Jung and Stolle 1981; Ramey et al. 1987). Several opposing factors influence the magnitude of the initial response. Physiological factors such as the area of cold stimulation, lung volume, alterations of the intrathoracic pressure (Müller's and Valsalva's manoeuvres), level of physical activity, age, long and short-term training, and psychological factors such as anxiety have been shown to influence the magnitude of apnoeic bradycardia in humans (Gooden 1982; Hentsch and Ulmer 1984).

In diving mammals there appears to be a species specific effect of water temperature on the diving response. While a positive correlation between water temperature and diving bradycardia has been found in the muskrat (Thornton et al. 1978), the bradycardia of the harbour seal has not appeared to be influenced by water temperatures over the same range (Dykes 1974).

Many studies have shown an inverse relationship between the extent of human diving bradycardia and water temperature at immersion (Craig 1963; Andersen 1966; Kawakami et al. 1967; Corriol and Rohner 1968; Song et al. 1969; Moore et al. 1972; Speck and Bruce 1978; Furedy and Morrison 1983; Paulev et al. 1990), while other studies have found no such correlation (Folinsbee 1974; Hunt et al. 1975; Hayward et al. 1984; Gallo et al. 1988). Often the ambient air temperature has not been reported in these studies. Our hypothesis is that ambient temperature has a significant effect on the bradycardia at apnoea or apnoeic facial immersion. We believe that this may account for some of the discrepancies among the results previously reported from different studies on the influence of water temperature. The present study has been designed to elucidate the influence of both water and ambient air temperatures on human diving bradycardia.

Methods

Subjects

A group of 23 healthy students, 11 women and 12 men, ranging in age from 21 to 33 years (mean 25 years) participated in this study. None of the subjects was experienced in apnoeic diving, but 5 had attained high levels of physical fitness through participating in other sports, 2 by swimming. Only 2 subjects were occasional smokers.

Protocol

The experiments were designed in consultation with a psychologist and a physician, and complied with the principles of the Helsinki

Declaration. The volunteers were thoroughly informed about the experimental procedures, and gave their consent to participate. All 23 subjects performed the following series of tests: after 60 min of exposure to air temperatures at 10, 20 or 30°C, sitting in a temperature controlled room, the subjects performed one apnoea in air and a series of apnoeas with the face immersed in water at 10, 20 or 30°C. The series of apnoeas was performed once in each air temperature on 3 subsequent days. The order of the water and air temperatures was randomized for each subject. In mammalian divers it has been shown that psychological factors, probably including a determination to perform a long dive, influence the magnitude of bradycardia (Kooyman 1989). As this may be the case also in humans, we asked the subjects to perform maximal effort apnoeic episodes in all the tests. The subjects wore optional indoor clothing during exposure and the test series. The same type of clothing was used for all the temperatures.

On 5 of the 23 subjects (three men and two women) oral and skin temperatures were measured from the start of exposure until the end of the test series. Oral and skin temperatures during exposure to 20°C ambient air were measured on an additional 5 subjects, who did not participate in the apnoea test series. The temperatures were recorded by thermocouples applied to the skin of the forehead, back of left hand, left thumb, back (between the shoulders), chest and in the mouth (according to Händel and Holmer 1986). The thermocouples were connected to a logger (Swema-logg 15, Swema Sweden) and a portable PC (Toshiba PC 1200).

Test series

The subject relaxed, lying prone on a mattress with the head on a rigid pillow under which a water container was placed, until a stable resting HR was established (Fig. 1a). The subject was instructed to hold his/her breath without prior hyperventilation after a deep but not maximal inspiration, and to avoid Müller's and Valsalva's manoeuvres during breath-holding. At a signal, the subject raised his/her head, the pillow was removed and after count-down the subject immersed the face in the water and held the breath for the maximal time (Fig. 1b–c). During apnoea in air, the same manoeuvres were performed, but without immersion of the face (Fig. 1a–b).

The apnoeic episodes were spaced apart by 8 to 10 min of rest, allowing the subject to recover to a stable resting HR. Water temperatures were maintained within $\pm 1^\circ\text{C}$ by the addition of ice

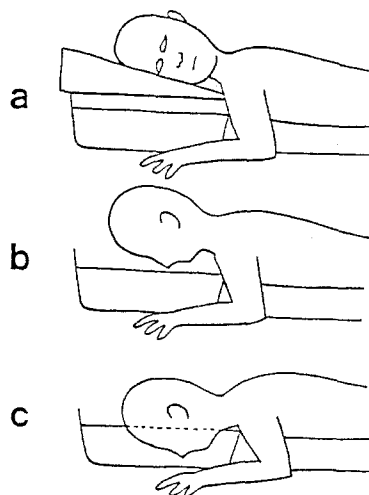


Fig. 1 Positions taken by the subject during apnoea in air (a–b), and apnoea with face immersion (a–c)

or hot water, and air temperatures were maintained within $\pm 1.5^{\circ}\text{C}$ with a thermostat throughout the exposures.

An electrocardiogram was recorded (with a Cardysuny 501 SA, lead II) from 30 s prior to each apnoeic episode and continuously until 30 s after the subject had resumed breathing. The 30-s period with resting HR preceding apnoea/apnoeic immersion was used as a reference, and the last 10 s of each apnoeic episode was analysed for bradycardia.

For the statistical evaluation of inter and intra air temperature response variation, a paired student's *t*-test with correction of the levels of significance for multiple comparisons was used. For the evaluation of the general effects of water and air temperature, ANOVA statistics with multiple comparison were carried out. The individual variations were examined and the results are presented as least square means with standard errors. A *P* less than 0.05 was considered to be statistically significant.

Results

Exposure to air temperatures

Temperature measurements on the subjects during exposure to air at 10, 20 or 30°C , revealed that skin temperatures were affected by the temperature of the ambient air, while oral temperatures remained unchanged after 1 h (Fig. 2). The temperature difference between cold (10°C) and warm (30°C) ambient air was within 5°C on the chest, back and forehead, while on the hand and thumb the difference was 15 and 20°C , respectively. Exposure to different ambient air temperatures affected the resting HR (reference HR). The HR in 10°C ambient air was significantly lower than that observed in 30°C air (Fig. 3).

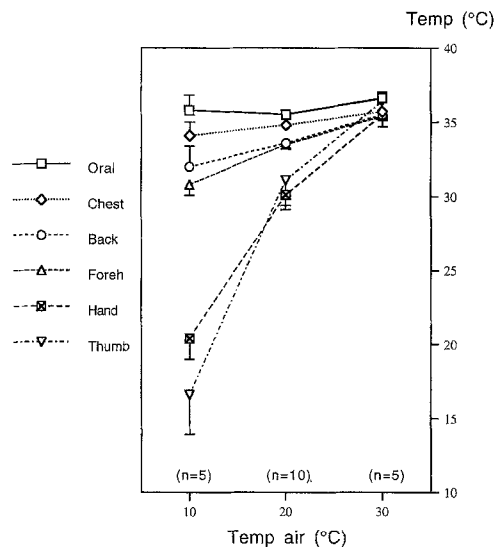


Fig. 2 Mean oral and skin temperatures after 60 min of exposure to different ambient air temperatures; means with SEM for five and in the 20°C condition for ten subjects

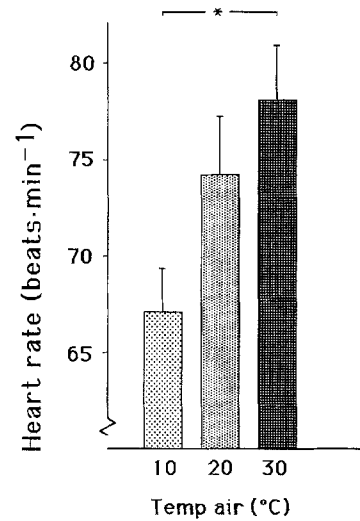


Fig. 3 Heart rates after exposure to different ambient air temperatures; means with SEM for 23 subjects. Significant differences ($P < 0.05$) are indicated by *

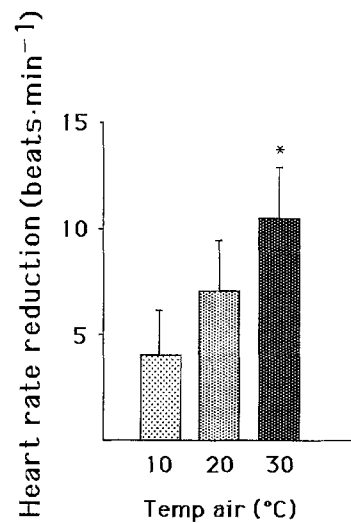


Fig. 4 Heart rate reductions during apnoea in air after exposure to different ambient air temperatures; means with SEM for 23 subjects. Bradycardia significantly different ($P < 0.05$) from reference HR is indicated by *

Series of apnoeas

All the subjects could hold their breath for at least 40 s, sufficiently long for a diving bradycardia to be established. At all ambient air temperatures, apnoea resulted in bradycardia, which was enhanced by water immersion (Figs. 4, 5). No significant sex differences were found in the data but the individual variation was large. There were 2 subjects who had significantly greater mean bradycardias than the other subjects, but showed the same response pattern as the group in general. All data are therefore presented for the 23 subjects as a group.

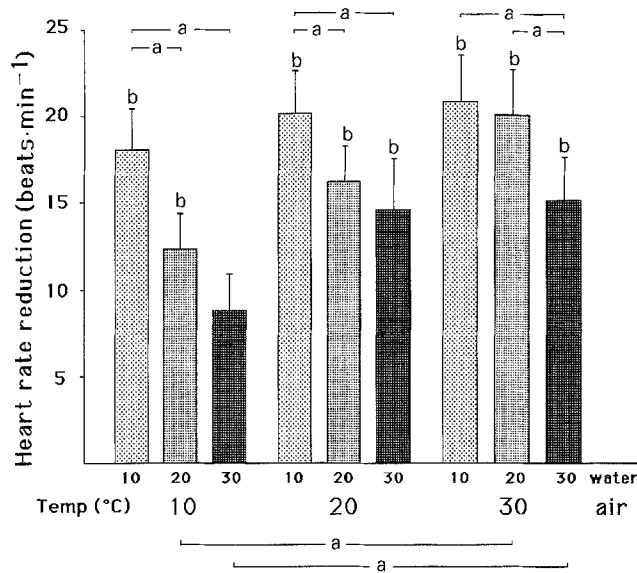


Fig. 5 Heart rate reductions during apnoea with face immersion in water of different temperatures after exposure to different ambient air temperatures; means with SEM for 23 subjects. Significance ($P < 0.05$) between the different air-water combinations is indicated by a. Bradycardia significantly different from reference HR is indicated by b

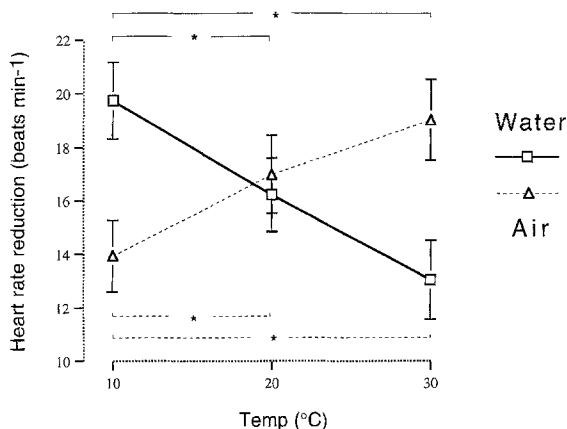


Fig. 6 The response to air temperature, when all water temperatures are pooled together, and the response to water temperature with all air temperatures pooled together; least square means with SEM for 23 subjects. Significance ($P < 0.05$) is indicated by *

The general effect of air and water temperatures

Both air and water temperatures significantly affected immersion bradycardia. An increase in air temperature in general increased bradycardia, while an increase in water temperature decreased bradycardia (Fig. 6). The responses observed at low air and water temperatures (10°C) differed significantly from those found at the higher temperatures (20 and 30°C).

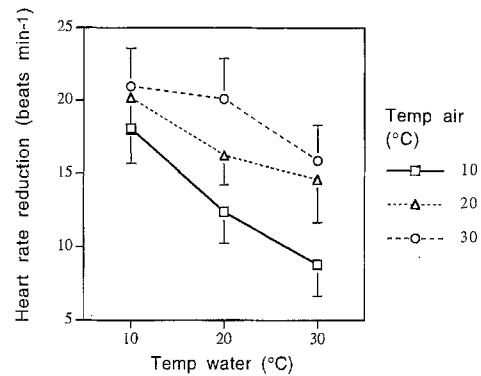


Fig. 7 Response to water temperature in 10, 20 and 30°C ambient air

The effect of air temperature on the response to apnoea

There were no significant differences among the responses to apnoea at the various ambient air temperatures (Fig. 4). Only in 30°C ambient air did the bradycardia differ significantly from the reference HR.

The effect of air temperature on the response to apnoea with immersion

When the responses to each air and water temperature were treated individually, the most pronounced immersion bradycardia was found at 30°C air and 10°C water immersion. The weakest response was observed at 10°C air and 30°C water immersion (Fig. 5), when the bradycardia was not significantly different from the reference HR. All other immersions resulted in significant bradycardia.

Exposure to different ambient air temperatures affected the patterns of response to water temperatures (Fig. 7). At 10 and 20°C ambient air there was a significant effect of changing the water temperature from 10 to 20°C, but not from 20 to 30°C. In 30°C air there was a significant effect of changing the water temperature from 20 to 30°C, but not from 10 to 20°C (Figs. 5, 7).

Discussion

Exposure to air temperatures

The differences in skin temperatures observed between cold (10°C) and normal room (20°C) air temperatures indicated that there was a cold-induced vasoconstriction in the cold air at the start of the test. This was also reflected by the decrease in HR, mediated by baroreceptors, preventing blood pressure from increasing in the cold air. In the warm (30°C) room, the high skin temperatures noted showed that the subjects reacted with vasodilatation in this situation, leading to

a higher HR to maintain blood pressure. It would appear that circulatory adaptations are sufficient means of thermoregulation in this range of ambient temperatures, since no changes in oral temperature were detected and no shivering thermogenesis was observed.

The effect of ambient air and water temperature on apnoea and apnoeic immersion bradycardia

The results of the present study are in agreement with earlier reports on the presence of human diving bradycardia triggered by apnoea and chilling of the face. Both ambient air temperature and water temperature are important in determining the extent of diving bradycardia in humans. The general difference in magnitude of bradycardia among different ambient air temperatures is in accordance with the findings of Moore et al. (1972); their data resemble the situation at 10 and 20°C ambient air in our study.

However, an additional effect of ambient temperature was revealed in the present study. The pattern of response to different water temperatures changed when the ambient temperature was increased to 30°C. It appeared that the bradycardia was dependent on water temperature within a range determined by the ambient air temperature. In the data presented by Corriol and Rohner (1968), Speck and Bruce (1978) and Paulev et al. (1990) a partial correlation to water temperature could be seen: in 20–25°C ambient air, immersion bradycardia was negatively correlated to water temperatures over 5–15°C, while there was little difference when water temperature varied between 20 and 30°C. This agrees with our observations for 10 and 20°C ambient air, but our study indicated that there may be a different effect of water temperature in 30°C ambient air.

The mechanisms of temperature dependence

We suggest that the two main physiological mechanisms responsible for the effects of environmental temperature on the magnitude of bradycardia are, firstly, the increase in vasodilatation (and HR) with increasing ambient temperature. Thus any apnoea or apnoeic immersion will result in a more pronounced vasoconstriction and, due to the baroreflex, a larger bradycardia the higher the ambient temperature. Secondly, the high dynamic sensitivity of the cutaneous thermoreceptors (Hensel 1973) would appear to play a role. While it has been found that the static cold-receptor response involves constant discharge at a given temperature within fixed limits, the dynamic cold-receptor response depends on rapid temperature change, the size and rate of change determining its magnitude (Hensel 1963). This implies that the size of the response at a given cold stimulus would be determined in part by the ambient temperature before stimulation.

At apnoeic facial immersion, the magnitude of the HR response would thus be a function of both ambient air and water temperature. The present data suggest that, depending on the initial temperature, the minimal and maximal stimuli occur at different water temperatures. After exposure to air at 10 or 20°C, a small response was elicited on immersion in water at 20°C, and the response remained the same when the water temperature was increased to 30°C. In ambient air at 30°C, the maximal response appeared to be elicited at 20°C water temperature, as there was no increase in the response when the colder water was used (Figs. 4, 7). Also in the data from Corriol and Rohner (1968) the effect of a 5°C decrease in the water temperature has been shown to be reduced when approaching 0°C. Beyond the upper and lower temperature limits it appears that little discrimination between temperatures can be made, which results in HR responses of equal magnitude. Thus there may be a sigmoid response pattern of HR to water temperature during apnoea with facial chilling.

Several studies (Folinsbee 1974; Hunt et al. 1975; Hayward et al. 1984; Gallo et al. 1988), have reported a lack of water temperature dependence. This may indicate that the water temperatures tested were all in the range compared to the ambient air temperature either above or below the interval where the response varies with water temperature. As has been pointed out by Paulev et al. (1990), the conclusion made by Gallo et al. (1988) may be questioned: when testing face-immersion in water at 5, 15 and 25°C at a room temperature of 25°C, their results indicated a significantly more pronounced bradycardia at 5°C compared to 15 and 25°C after the initial period of HR decline. This is in accordance with the results presented here, resembling the situation in ambient air at 20°C.

Thus, a difference in ambient air temperature may account for the different conclusions in previous reports on the influence of water temperature on human diving bradycardia. Therefore we recommend that ambient air temperature should always be reported in studies of the human diving response. In animal studies, the possible effect of ambient air temperatures should also be considered, so that the effect of water temperature will be tested at the appropriate ambient temperature for the species being investigated.

The temperature dependence of the human diving response has been used as an argument against its efficiency in actual apnoeic diving situations, which usually occur in warm waters, and against the proposition by Hardy (1977) that it may be a relic of an adaptation to diving activities acquired during a semi-aquatic phase in hominid evolution (Mukhtar and Patrick 1986). However, the present findings suggest that the diving response may very well be operative in a tropical diver, as long as his/her face is exposed between dives to air sufficiently warmer than the water (Schagatay 1991).

Acknowledgements This investigation was supported by grants from the Hierta Retzius Foundation, the Royal Physiographic Society, Lund, and the Faculty of Science, University of Lund. We are also grateful to all the volunteers who participated in the experiments.

References

- Andersen HT (1966) Physiological adaptations in diving vertebrates. *Physiol Rev* 46:212–224
- Butler PJ, Woakes AJ (1987) Heart rate in humans during underwater swimming with and without breath-hold. *Respir Physiol* 69:387–399
- Corriol J, Rohner JJ (1968) Rôle de la température de l'eau dans la bradycardie d'immersion de la face. *Arch Sci Physiol* 22:265–274
- Craig AB Jr (1963) Heart response to apneic underwater diving and to breath-holding in man. *J Appl Physiol* 18:854–862
- Dykes RW (1974) Factors related to the dive reflex in harbour seals: respiration, immersion bradycardia and lability of the heart rate. *Can J Physiol Pharmacol* 52:259–262
- Elsner R, Gooden B (1983) Diving and asphyxia: a comparative study of animals and man. *Physiological Society monograph* 40. Cambridge University Press, Cambridge, p. 175
- Folinsbee L (1974) Cardiovascular response to apneic immersion in cool and warm water. *J Appl Physiol* 36:226–232
- Furedy JJ, Morrison JW (1983) Effects of water temperature on some noninvasively measured components of the human dive reflex: an experimental response-topography analysis. *Psychophysiology* 20:569–578
- Gallo L Jr, Maciel BC, Manco JC, Marin Neto JA (1988) Limitations of facial immersion as a test of parasympathetic activity in man. *J Physiol* 396:1–10
- Gooden BA (1982) The diving response in clinical medicine. *Aviat Space Environ Med* 53:273–276
- Hardy A (1977) Was there a *Homo aquaticus*? *Oxford University Scientific Society, Zenith* 15:4–6
- Hayward JS, Hay C, Matthews BR, Overweel CH, Radford DD (1984) Temperature effect on the human dive response in relation to cold water near-drowning. *J Appl Physiol Respir Environ Exerc Physiol* 56:202–206
- Hensel H (1963) Electrophysiology of thermosensitive nerve endings. In: Herzfeld CM (ed) *Temperature, its measurement and control in science and industry 3: biology and medicine*. Reinhold, New York, pp 191–198
- Hensel H (1973) Cutaneous thermoreceptors. In: Iggo A (ed) *Handbook of sensory physiology*, vol. II. Springer, Berlin Heidelberg, New York, pp 79–110
- Hentsch U, Ulmer H-V (1984) Trainability of underwater breath-holding time. *Int J Sport Med* 5:343–347
- Hunt NG, Whitaker DK, Willmott NJ (1975) Water temperature and the “diving reflex”. *Lancet* I:572
- Hurwitz BE, Furedy JJ (1986) The human dive reflex: an experimental, topographical and physiological analysis. *Physiol Behav* 36:287–294
- Händel SE, Holmér I (1986) Climate, clothes and textiles (in Swedish). Contact, (STU-information), Stockholm
- Jung K, Stolle W (1981) Behaviour of heart rate and incidence of arrhythmia in swimming and diving. *Biotelem Patient Monit* 8:228–239
- Kawakami Y, Natelson BH, DuBois AB (1967) Cardiovascular effects of face immersion and factors affecting diving reflex in man. *J Appl Physiol* 23:964–970
- Kooyman GL (1989) Diverse divers. *Physiology and behaviour. Zoophysiology*, Vol. 23. Springer, Berlin Heidelberg New York, p. 200
- Moore TO, Lin YC, Lally DA, Hong SK (1972) Effects of temperature, immersion and ambient temperature on human apneic bradycardia. *J Appl Physiol* 33:36–41
- Mukhtar MR, Patrick JM (1986) Ventilatory drive during face immersion in man. *J Physiol* 370:13–24
- Paulev P-E, Pokorski M, Honda Y, Ahn B, Masuda A, Kobayashi T, Nishibayashi Y, Sakakibara Y, Tanaka M, Nakamura W (1990) Facial cold receptors and the survival reflex “diving bradycardia” in man. *Jpn J Physiol* 40:701–712
- Ramey CA, Ramey DN, Hayward JS (1987) Dive response of children in relation to cold-water near-drowning. *J Appl Physiol* 63:665–668
- Schagatay E (1991) The significance of the human diving reflex. In: Roede M, Wind J, Patrick J, Reynolds V (eds) *The aquatic ape: fact or fiction?* Souvenir Press, London, pp 247–254
- Scholander PF (1940) Experimental investigations on the respiratory function in diving mammals and birds. *Hvalradets Skr Nor Vidensk Akad Oslo* 22:1–131
- Schuitema KE, Holm B (1988) The role of different facial areas in eliciting human diving bradycardia. *Acta Physiol Scand* 132:119–120
- Song SH, Lee WK, Chung YA, Hong SK (1969) Mechanisms of apneic bradycardia in man. *J Appl Physiol* 27:323–327
- Speck DF, Bruce DS (1978) Effects of varying thermal and apnoeic conditions on the human diving reflex. *Undersea Biomed Res* 5:9–14
- Thornton R, Gordon C, Ferguson JH (1978) Role of thermal stimuli in the diving response of the muskrat *Ondatra zibethica*. *Comp Biochem Physiol* 61 A:369–370